An update to "On the sign patterns of entrywise positivity preservers in fixed dimension"

Terence Tao · 30 September 2017

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Apoorva Khare and I have updated our paper "On the sign patterns of entrywise positivity preservers in fixed dimension", announced at this post from last month. The quantitative results are now sharpened using a new monotonicity property of ratios $s_\lambda(u)/s_\mu(u)$ of Schur polynomials, namely that such ratios are monotone non-decreasing in each coordinate of u if u is in the positive orthant, and the partition λ is larger than that of μ. (This monotonicity was also independently observed by Rachid Ait-Haddou, using the theory of blossoms.) In the revised version of the paper we give two proofs of this monotonicity. The first relies on a deep positivity result of Lam, Postnikov, and Pylyavskyy, which uses a representation-theoretic positivity result of Haiman to show that the polynomial combination

$$s_{(\lambda\wedge\nu)/(\mu\wedge\rho)} s_{(\lambda\vee\nu)/(\mu\vee\rho)} - s_{\lambda/\mu} s_{\nu/\rho} \tag{1}$$

of skew-Schur polynomials is Schur-positive for any partitions λ, μ, ν, ρ (using the convention that the skew-Schur polynomial $s_{\lambda/\mu}$ vanishes if μ is not contained in λ, and where $\lambda \wedge \nu$ and $\lambda \vee \nu$ denotes the pointwise min and max of λ and ν respectively). It is fairly easy to derive the monotonicity of $s_\lambda(u)/s_\mu(u)$ from this, by using the expansion

$$s_\lambda(u_1, \dots, u_n) = \sum_k u_1^k s_{\lambda/(k)}(u_2, \dots, u_n)$$

of Schur polynomials into skew-Schur polynomials (as was done in this previous post).

The second proof of monotonicity avoids representation theory by a more elementary argument establishing the weaker claim that the above expression (1) is non-negative on the positive orthant. In fact we prove a more general determinantal log-supermodularity claim which may be of independent interest:

> **Theorem 1** Let A be any $n \times n$ totally positive matrix (thus, every minor has a non-negative determinant). Then for any k-tuples I_1, I_2, J_1, J_2 of increasing elements of $\{1, \dots, n\}$, one has
>
> $$\det(A_{I_1\wedge I_2, J_1\wedge J_2}) \det(A_{I_1\vee I_2, J_1\vee J_2}) - \det(A_{I_1,J_1}) \det(A_{I_2,J_2}) \geq 0$$
>
> where $A_{I,J}$ denotes the $k \times k$ minor formed from the rows in I and columns in J.

For instance, if A is the matrix

$$A = \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix}$$

for some real numbers $a, \dots, p$, one has

$$ah - de \geq 0$$

(corresponding to the case $k = 1$, $I_1 = (1), I_2 = (2), J_1 = (4), J_2 = (1)$), or

$$\det\begin{pmatrix} a & c \\ i & k \end{pmatrix} \det\begin{pmatrix} f & h \\ n & p \end{pmatrix} - \det\begin{pmatrix} e & h \\ i & l \end{pmatrix} \det\begin{pmatrix} b & c \\ n & o \end{pmatrix} \geq 0$$

(corresponding to the case $k = 2$, $I_1 = (2,3)$, $I_2 = (1,4)$, $J_1 = (1,4)$, $J_2 = (2,3)$). It turns out that this claim can be proven relatively easy by an induction argument, relying on the Dodgson and Karlin identities from this previous post; the difficulties are largely notational in nature. Combining this result with the Jacobi-Trudi identity for skew-Schur polynomials (discussed in this previous post) gives the non-negativity of (1); it can also be used to directly establish the monotonicity of ratios $s_\lambda(u)/s_\mu(u)$ by applying the theorem to a generalised Vandermonde matrix.

(Log-supermodularity also arises as the natural hypothesis for the FKG inequality, though I do not know of any interesting application of the FKG inequality in this current setting.)

Comments (9)

Fred Lunnon · 30 September 2017

Formulae preceding “corresponding to the case {k=1}” and
“corresponding to the case {k=2}” should read ” ... >= 0 ” ?
WFL
[Corrected, thanks – T.]

Suvrit Sra (@optiML) · 1 October 2017

A minor clarification regarding page 35 of your paper. It says that “the conjecture – as stated in [8] and proved in [34] – has a minor error, namely when $\sum_j m_j > \sum_j n_j$” — In [34] the conjecture is proved for true majorization, where $\sum_j m_j = \sum_j n_j$ holds. Or am I missing something? Thanks!

> **Terence Tao** · 2 October 2017
>
> Thanks for this, it looks like we did not quite quote the CGS conjecture correctly, and will sort this out in the next revision of the ms.

Rachid Ait-Haddou · 1 October 2017

Dear Terence,
Will it be OK to change my first name to Rachid instead of Rachit :). Many thanks for the mention. Also as Sra just mentioned, what is proved in the paper is slightly different from Cuttler-Greene-Skandera conjecture. The paper proves that if $m_i \geq n_i$ for all i then the inequality between the normalized Schur functions holds. The conjecture states that if $\sum_{k=1}^{j} m_k \geq \sum_{k=1}^{j} n_k, j = 1, 2, ..., N-2$ and $\sum_{k=1}^{N-1} m_k = \sum_{k=1}^{N-1} n_k$ then the inequality between the normalized Schur functions holds.

> **Terence Tao** · 2 October 2017
>
> Sorry for the typo! We’ll correct the attribution to the CGS paper in the next revision of the ms.

Rachid Ait-Haddou · 1 October 2017

It is sometimes confusing to go from powers to partitions (when it is done hazily). I would like to correct the statement of the conjecture in terms of the powers. It states that if $\sum_{k=j}^{N-1} m_k \geq \sum_{k=j}^{N-1} n_k, j = 1, 2, ..., N-1$ and $\sum_{k=0}^{N-1} m_k = \sum_{k=0}^{N-1} n_k$ then the inequality between the associated normalized Schur functions holds over $]0, \infty[^N$. I hope I got it right this time.

Aula · 2 October 2017

How does Theorem 1 say anything about the value of the expression ah-dg? If I understood the definitions correctly, it should probably be ah-de instead.
[Corrected, thanks -T.]

A. Barreras · 16 October 2017

Dear Terence,
In Linear Algebra literature, there exist two “school of notations”:
1: TP: matrices with non-negative minors
STP: matrices with strictly positive minors
2: TN: matrices with non-negative minors
TP: matrices with posive minors
If I understood correctly, you are using the first one. In this case, in the definition of totally positive (p. 38 of the article, line -3), should be changed formula “det(A_{I,J})>0” by “det(A_{I,J}) \geq 0”?

Sorry if this is just my misunderstood.

1. Allan Pinkus (2009), Totally Positive Matrices, Cambridge University Press, ISBN 9780521194082
2. Fallat, S. M., & Johnson, C. R. (2011). Totally nonnegative matrices. Princeton University Press.



Terence Tao · 17 October 2017

In our paper we're using total positivity in the strict sense. (A large part of the paper is concerned instead with positive semi-definite matrices, though.)